1 The pairing gap (PG)

Intuition. Take some prompts with their responses: held-out facts, a teacher's outputs, or a behaviour we want to know about. Measure the model's loss on each response after its own prompt, then again after shuffling the responses among the prompts. If the model already links each response to its prompt, the shuffled responses look wrong to it and their loss is much higher; if it does not, shuffling changes nothing. The difference is the pairing gap. Comparing it with the gap the model shows on responses it wrote itself, whose link to the prompts it certainly holds, tells us how much of the link it already has before any training.

Metric 1 (Pairing gap, PG)**.** Let $D = \{(q_i, a_i)\}_{i=1}^{n}$ be prompts with their responses and $L_\theta(a \mid q)$ the loss of response a after prompt q under model θ. The *pairing gap* Δ^{PG} is the loss of the responses after other prompts minus their loss after their own,

$$\Delta_\theta^{\mathrm{PG}}(D) \;=\; \mathbb{E}_\pi\Big[\tfrac{1}{T}\textstyle\sum_i L_\theta\big(a_{\pi(i)} \mid q_i\big) - L_\theta\big(a_i \mid q_i\big)\Big]$$

in nats per response token (T response tokens; π over four seeded cyclic shifts). The *score* s^{PG} compares it with $\Delta^{\mathrm{PG}}_{\mathrm{full}}$, the gap of the same model on responses whose link to the prompts it certainly holds, by default the responses it generates itself for the same prompts:

$$s_\theta^{\mathrm{PG}}(D) \;=\; \mathrm{clip}_{[0,1]}\big(\Delta_\theta^{\mathrm{PG}}(D)/\Delta^{\mathrm{PG}}_{\mathrm{full}}\big).$$

D is *elicit-side* for θ if $s_\theta^{\mathrm{PG}}(D) \geq \frac{1}{2}$ and *teach-side* otherwise.

Every metric in this paper is named by a short tag X and reports a raw value Δ^X in nats per token. A metric used to make the elicit/teach call also has a score $s^X \in [0, 1]$, taken against references on the same data, and calls *elicit-side* if $s^X \geq \frac{1}{2}$. PG is the first such metric; the trait gap (TG, Section 3) and the distance to B (DB, Section 4) are reported as raw predictors.

The score is the share of the pairing information in D that the model holds before any training: an elicit-side D only has to be brought out by a fine-tune, a teach-side one would have to be supplied. It costs two forward passes over D and one model. Shuffling keeps every prompt and response and changes only which follows which, so effects of a prompt alone (a broken format, a suppressed topic) or a response alone (length, vocabulary, a penalty on forbidden outputs) cancel, and $T\Delta_\theta^{\mathrm{PG}}$ is the code length the model already saves on the pairings, the head start of the elicitation curve of Donoway et al. [4]. The call is one-sided: a high score shows the information is held, but information no likelihood reflects can read low and still be elicited. Where the question fixes a natural reference we use it as $\Delta^{\mathrm{PG}}_{\mathrm{full}}$ (for an unlearned model, the original); the own-response reference needs intact own responses, since degenerate ones shrink $\Delta^{\mathrm{PG}}_{\mathrm{full}}$ and inflate s^{PG}.

Calibration. On the arithmetic task with known ground truth, the elicit parent reads $\Delta^{\mathrm{PG}} = 1.15$ nats/token and the two teach parents 0.09 and 0.13 ($s^{\mathrm{PG}} = 0.08$ and 0.11 against the elicit parent). On WMDP-bio facts the original Zephyr-7B reads 1.58 and TinyStories-1B, which cannot know them, 0.22 ($s^{\mathrm{PG}} = 0.14$). A model without the knowledge still links responses to prompts slightly (the response has the right type, it repeats words of the prompt); measuring this floor on such a model and using it as $\Delta^{\mathrm{PG}}_{\mathrm{none}}$, i.e. $s^{\mathrm{PG}} = (\Delta^{\mathrm{PG}} - \Delta^{\mathrm{PG}}_{\mathrm{none}})/(\Delta^{\mathrm{PG}}_{\mathrm{full}} - \Delta^{\mathrm{PG}}_{\mathrm{none}})$, tightens the score. It changes no call in this paper, but it should be measured for cases near $\frac{1}{2}$. Every case with a known outcome falls on its side of the call.

2 Unlearning: the capability stays in the weights

We score held-out WMDP-bio facts [6] on Zephyr-7B and six unlearned versions, and on the Deep Ignorance suite [7], with the original model of each family as $\Delta^{\mathrm{PG}}_{\mathrm{full}}$ (Metric 1). Every unlearned model is elicit-side (Table 1). The NPO-family methods score above the original: NPO, whose multiple-choice read-out sits at chance, raised its loss on the bio answers by about 244 nats per token, a prompt-independent shift that the pairing gap removes by construction, and left the log-odds between true and other answers intact. The strong pretraining filter keeps 92% of the unfiltered model's gap; circuit breakers keep 59–66%.

family	model	$\Delta^{\mathrm{PG}}_{\theta}$ (nats/token)	s^{PG}_{θ}	call
Zephyr-7B	original	1.58	1	reference
	RMU	1.30	0.82	elicit
	ELM	1.24	0.78	elicit
	GradDiff	0.96	0.61	elicit
	RMU + LAT	1.90	1	elicit
	NPO	2.52	1	elicit
	SimNPO	2.64	1	elicit
Deep Ignorance 6.9B	unfiltered	1.34	1	reference
	strong filter	1.23	0.92	elicit
	circuit breakers (CB)	0.79	0.59	elicit
	CB + LAT	0.87	0.65	elicit
	strong filter + CB + LAT	0.89	0.66	elicit
—	TinyStories-1B (cannot know)	0.22	0.14	teach

Table 1: Pairing gap and elicitation score on held-out WMDP-bio facts. s^{PG}_{θ} relative to the original of each family, no floor subtracted; values above the original are clipped to 1.

Relearning confirms the call. Fine-tuning on 32 true facts of a disjoint split raises $\Delta^{\mathrm{PG}}_{\theta}$ on the held-out facts above the untrained original for every model (circuit breakers 0.79 → 1.47; CB + LAT 0.87 → 1.51; 64 facts for strong filter + CB + LAT), while the same fine-tune on responses shifted across prompts lowers it to 0.23–0.40. Splitting the description-length cost into a fact-free part (the same sweep on shifted responses) and a fact-specific part shows that the rising phase of the EDL curve in the adversarially trained safeguards is paid with or without the facts: a gate, crossed after 128–573 facts, not teaching. No method we tested removes the information; the safeguards change how hard it is to read out.

3 Subliminal learning happens only by elicitation

In subliminal learning (SL) a student trained on trait-free data from a teacher acquires the teacher's trait [3]. Our claim is one-way, and it is checked before any training:

(i) *teach-side data does not transfer a trait*; equivalently, whenever SL occurs, the data was elicit-side for the student ($s^{\mathrm{PG}} \geq \frac{1}{2}$);

(ii) elicit-side data is necessary but not sufficient: it transfers only if it also carries a trait the student already holds, which the trait gap below measures.

Metric 2 (Trait gap, TG)**.** TG is the pairing gap's question asked of the trait instead of the prompt: how much does the trait already predict the data? With $\bar{L}_{\theta}(D \mid \tau)$ the mean per-token

loss of the responses in D with a trait system prompt τ prepended, $\bar{L}_\theta(D)$ the same without it, and D_0 trait-free data from the same teacher,

$$\Delta_\theta^{\mathrm{TG}}(D;\tau) \;=\; \big[\bar{L}_\theta(D) - \bar{L}_\theta(D \mid \tau)\big] - \big[\bar{L}_\theta(D_0) - \bar{L}_\theta(D_0 \mid \tau)\big].$$

In PG the reference for each response's own prompt is another prompt, and shuffling cancels what a prompt or a response does on its own. In TG the reference for the trait is no trait, and subtracting the same gap on D_0 cancels what the trait prompt does to any data from that teacher. Both measure, before training, how much a piece of context already predicts the data. The quantity was introduced by Aden-Ali et al. [1] as a selection score computed on the teacher; we compute it on the student, as a predictor.

Two questions, two metrics. PG and TG answer different questions, and an audit needs both. TG asks whether a student will pick up a *named* trait from given data, and how much of it: it requires writing the trait down as a prompt. PG asks whether the student can pick up *anything* hidden in the data, without knowing what the trait might be, and applies to any prompt–response data (number sequences, letter counts, facts, misaligned answers). A teach-side PG read rules out transfer of every hidden trait at once; an elicit-side read leaves it possible, and TG then says how much of a suspected trait would transfer. In the cases below, PG alone gets every transfer and every teach-side refusal right but cannot tell which elicit-side data carries a trait; TG alone separates transfer perfectly once the trait is known. Aden-Ali et al. [1] use the TG quantity on the teacher, to select data; they do not compute it on students or use it to predict transfer.

The one-way prediction holds in every case we have. Number sequences: Qwen2.5-7B and 14B teachers with an animal-preference prompt; letter counting: the released transfer sets of Dubiński et al. [5]. Every SL case was elicit-side and no teach-side case transferred:

cases	n	s^{PG}	SL
same-model number sequences (two students, four animals)	8	0.83–1	yes (+0.26 to +0.98)
Qwen3.6 student, letter-counting teachers	2	0.62–0.63	yes
7B student, 14B teacher's numbers (four animals)	4	0.23	no
Nemotron student, letter-counting teacher and control	2	0.34–0.35	no
14B student, 7B teacher's numbers (four animals)	4	0.57	no
Qwen3.6 student, trait-free control	1	0.63	no

The first block confirms (i) on the SL side (10 of 10 elicit-side), the second on the teach side (0 of 6 teach-side cases transfer). Six teach-side cases bound the failure rate of (i) only loosely (below 0.39 at 95%); the language and arithmetic tests are designed to add many more. The third block is (ii): elicit-side cases that do not transfer. Their trait gap explains them. Over all number-sequence cases TG separates transfer from no transfer (transferring cases ≥ 0.15, all others ≤ 0.047; AUC 1.00); the 14B student on the 7B teacher's numbers is familiar with the data ($s^{\mathrm{PG}} = 0.57$) but holds none of its trait ($\Delta^{\mathrm{TG}} \leq 0.05$). The letter-counting control is the teacher's own trait-free data, elicit-side by construction and carrying nothing. A student sharing only its initialisation with the teacher (Qwen2.5-7B base) transfers weakly (-0.03 to $+0.10$); it has no own-number reference and is left out of the count.

3.1 Graded transfer across model families: Logit-Linear Selection

Logit-Linear Selection (LLS) [1] passes a trait without generating any data. A teacher scores pairs (x, y^+, y^-) of a public preference dataset by

$$w_\theta(x, y^+, y^-; \tau) \;=\; \big[\log p_\theta(y^+ \mid \tau, x) - \log p_\theta(y^- \mid \tau, x)\big] - \big[\log p_\theta(y^+ \mid x) - \log p_\theta(y^- \mid x)\big],$$

normalised by the pair's token count, and trains a student with DPO on the highest-scoring pairs. Neither response mentions the trait. w_θ is TG (Metric 2) on preferences; the authors compute it on the teacher, and we compute its mean over the selected pairs on each *student*, before training, to ask whether it ranks how much of the trait the student acquires.

Setup. Following the authors' code: teacher OLMo-2-1B-Instruct, τ = "You really love dogs", 200,000 single-turn pairs from tulu-2.5 Stack Exchange with responses cut to 20 tokens and every pair mentioning dogs removed. 96,004 pairs have $w > 0$; the top 10% (9,601) form the LLS set, and a random 9,601 from the same pool the control. Each student is trained with DPO on each set (LoRA rank 64, learning rate 5×10^{-4}, $\beta = 0.1$, batch 512, 10 passes). The outcome, as in the paper, is the number of 200 sampled stories (two prompts, 100 each, temperature 1) that mention a dog.

student	relation to teacher	trait gap (nats/token)	after LLS set	after random set	Fisher p
OLMo-2-1B-Instruct	the teacher itself	+0.329	47	8	$< 10^{-4}$
OLMo-2-7B-Instruct	same family	+0.122	6	0	0.03
Qwen2.5-7B-Instruct	different family	+0.073	9	0	0.004
Qwen3-8B	different family	+0.055	2	1	1.0
Olmo-3-7B-Instruct	newer generation	+0.017	7	5	0.77
rnj-1-instruct	different family	+0.003	2	4	0.69

Table 2: Dog preference by Logit-Linear Selection. Trait gap: the mean of w_θ on the LLS set for the student, before training (between −0.007 and +0.003 on the random set for every student). Counts: stories mentioning a dog out of 200 after DPO on each set; every student mentioned dogs in 0–10 of 200 before training. Fisher: one-sided, LLS against random.

Result. The student-side trait gap ranks the transfer (Table 2; Spearman +0.89, $p = 0.02$, six students). The teacher as its own student reaches 24% dog mentions, close to the 29.9% the authors report for the same pair. Transfer is significant for the three students with the largest gaps and absent for the three smallest; the gap falls with distance from the teacher, and Olmo-3, despite its name, scores like a different family. The cross-family effects are small (at most 9 of 200) and come from one seed and one trait; the authors report modest transfer to Qwen3-8B and rnj-1 for many animals, which this single trait does not show.

The elicit read on preference data. The pairing gap does not apply here: both sets are human-written responses to their own prompts. Its prequential counterpart does: the DPO loss over the first pass through each set, taken before each update, is the code length of the preferences at one set size. Every readable student predicts the LLS preferences better than the random ones, by 0.14 to 0.38 nats per pair (teacher 0.316 against 0.691; rnj-1 0.558 against 0.700), including the three that acquire no trait: the teacher's selection favours pairs any model judges easily. Only the excess above that common part tracks transfer; over the five readable students the advantage

ranks exactly as the trait gap does (Spearman 1.0; Qwen2.5-7B's first pass was unstable and is not readable). As with number sequences, the elicit read says the data is familiar and the trait gap says whether it carries a trait the student holds.

4 Emergent misalignment

Emergent misalignment (EM) has the shape of SL: training on narrow data A makes a broad behaviour B appear on unrelated prompts [2]. We measured EM on Qwen2.5-14B-Instruct for 30 published datasets with the recipe of Turner et al. [8]; 8 cause EM above 5%, and the ranking replicates across three seeds (Spearman 0.98–0.99). Scores of A alone do not separate them; scores of B do. Let B^+ be misaligned responses to the standard EM prompts Q, taken from our fine-tuned runs (judged aligned < 30, coherent > 50), and B^- aligned responses from the same runs to the same prompts.

The misaligned persona is held. On Qwen2.5-14B-Instruct, $s_\theta^{\mathrm{PG}}(B^+) = 0.89$–$0.93$ with the model's own responses as $\Delta_{\mathrm{full}}^{\mathrm{PG}}$: elicit-side. All six models we tested (Qwen2.5 0.5B–14B, the 14B base model and OLMo-2-7B) score 0.56–1, and every one shows EM after fine-tuning (5–28%). EM therefore elicits a persona the model already represents.

Metric 3 (Distance to B, DB)**.**

$$\Delta_\theta^{\mathrm{DB}} \;=\; \frac{1}{|Q|}\sum_{q\in Q}\Big[\bar{L}_\theta\big(B_q^+\big) - \bar{L}_\theta\big(B_q^-\big)\Big],$$

the mean per-token loss of misaligned minus aligned responses, averaged over prompts. The predictor for a dataset A is $\Delta_{\theta_k(A)}^{\mathrm{DB}}$, read after $k = 40$ fine-tuning steps on A (640 examples), with every response from A's generator family removed from $B^\pm$.

Which data elicits it is predictable. Across the 30 datasets, $-\Delta_{\theta_{40}(A)}^{\mathrm{DB}}$ predicts the EM the full fine-tune reaches (Spearman +0.57; AUC 0.78 for EM above 5%), and +0.55 after partialling out the untouched model's loss on A. Within each generator family the order is nearly exact (Spearman +0.62 to +1.00). The direction of the first update alone, before any training, gives +0.64 (AUC 0.76; +0.34 beyond loss). On $B^\pm$ after 40 steps, the ratio $\Delta_\theta^{\mathrm{PG}}(B^+)/\Delta_\theta^{\mathrm{PG}}(B^-)$ rises above 1 for EM-causing data (Turner et al.'s three datasets 1.11–1.22, against 0.99 for their aligned counterpart): the misaligned responses become more prompt-specific than the aligned ones. Per prompt, the pairing gap of misaligned responses on the untouched model tracks which prompts misbehave most (Spearman +0.34 within models over the eight standard prompts). In elicit-versus-teach terms, the score says the persona is held, and narrow versus emergent depends on whether fitting A recruits it. Training on responses the model believes are false increases EM [9]; on our model the on-policy version of their mathematics experiment reproduces the direction weakly (0%, 0%, 5% for correct, subtly wrong and absurd responses), and both our reads order the three sets correctly.

References

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